

Course Code	Course Title							
116U01C401	Probability, Statistics and Optimization Techniques							
	TH		P		TUT		Total	
Teaching Scheme(Hrs.)	03		--		01*		04	
Credits Assigned	03		--		01		04	
Examination Scheme	Marks							
	CA		ESE	TW	O	P	P&O	Total
	ISE	IA						
	30	20	50	25	--	--	--	125

* Batch wise Tutorial

Course prerequisites:

- Basics of Statistics and Probability
- Introductory Linear programming problems

Course Objectives

This course Exposes students to the concepts of Correlation, Regression for given bivariate data. Students are made familiar with different discrete and continuous probability distributions. The course acquaints students with concepts of Large sample test, Small sample test and Chi – Square test. The course familiarizes students with different methods of solving Linear and Nonlinear Programming problems. Some basic queuing theory models are also discussed in the course. Using these methods it will be possible to analyze and interpret a given real life situation and think of possible solutions.

Course Outcomes

At the end of successful completion of the course the student will be able to

- CO1. Apply concepts of correlation, regression for given bivariate data.
- CO2. Apply concepts of Binomial, Poisson, Exponential and Normal distribution to solve Engineering problems.
- CO3. Apply Large sample test and small sample test to analyze collected data.
- CO4. Apply concepts of Linear and Nonlinear programming methods to solve problems.
- CO5. Apply the methods of single server limited queue and single server unlimited queue models to analyze and interpret the data.

Module No.	Unit No.	Details	Hrs.	CO
1	Probability and Probability Distribution		12	CO 1
	1.1	Conditional Probability, Bayes' theorem, Joint Probability		
	1.2	Discrete and Continuous Probability Distribution		
	1.3	Binomial Distribution, Poisson Distribution		
	1.4	Uniform Distribution, Normal Distribution, Exponential Distribution		
2	Correlation and Regression		06	CO 2
	2.1	Correlation, Co-variance, Karl Pearson Coefficient of Correlation & Spearman's Rank Correlation Coefficient.		
	2.2	Regression Coefficients, lines of regression & logistic regression		
3	Sampling Theory		07	CO 3
	3.1	Sampling distribution. Test of Hypothesis. Level of significance, critical region. One tailed and two tailed tests. Interval Estimation of population parameters. Large and small samples.		
	3.2	Difference between sample mean and population means for large samples, Test for significance of the difference between the means of two large samples.		
	3.3	Student's t-distribution: Test for significance of the difference between sample mean and population means, Test for significance of the difference between the means of two Samples, paired t-test.		
	3.4	Chi-square distribution as a Test of Independence, Test of the Goodness of fit and Yate's correction.		
	3.5	Fisher's z-test		
4	Optimization Techniques		13	CO 4
	4.1	Types of solution, Standard and Canonical form of LPP, Basic and feasible solutions, simplex method.		
	4.2	Artificial variables, Big –M method (method of penalty).		
	4.3	Duality and Dual Simplex method		

	4.4	Unconstrained optimization, problems of two or three variables with one equality constraint using Lagrange's Multiplier method.		
	4.5	Problems of two or three variables with one inequality constraint using Kuhn-Tucker conditions		
5	Queuing Theory		07	CO5
	5.1	Introduction, Features of Queuing, solution of Queuing models. M/M/1(Singal Server, Unlimited Queue Model)		
	5.2	M/M/1 Singal Server, limited Queue Model		
Total			45	

Recommended Books:

Sr. No.	Name/s of Author/s	Title of Book	Name of Publisher with country	Edition and Year of Publication
1.	B. S. Grewal	<i>Higher Engineering Mathematics</i>	Khanna Publications, India	43 rd Edition 2014
2.	Erwin Kreyszig	<i>Advanced Engineering Mathematics</i>	Wiley Eastern Limited, India	10 th Edition 2015
3.	J. K. Sharma	<i>Operation research: Theory and Applications</i>	Laxmi Publications, India	6 th Edition 2017
4.	S.C.Gupta and V.K.Kapoor	<i>Fundamentals of Mathematical Statistics</i>	Sultan Chand & Sons	11 th Edition 2009
5.	Ronald E. Walipole, Raymond H.Myers	<i>Probabilities & Statistics for Engineers & Scientists</i>	Pearson Education	9 th Edition 2010
6.	P. N. Wartikar and J. N. Wartikar	<i>A textbook of Applied Mathematics Vol I & II</i>	Pune Vidyarthigruha, India	6 th Edition 2012

Term-Work will consist of Tutorials covering the entire syllabus. Students will be graded based on continuous assessment of their term work. **At least 2 tutorials will be conducted with the help of Mathematical and Statistical software in the Laboratory.**